



REGISTRATION FORM MASTER RESEARCH PROJECT ASTRONOMY



STUDENT DATA

Student name Margarida Pólvara Fonseca

Student number s4851544

Specialisation Research in Astronomy

Email address dafonseca@strw.leidenuniv.nl

MASTER PROJECT

Project type Master (30 EC)

Starting date 03/09/2026 Final deadline 30/06/2027

Supervisor(s) Ana Monreal Ibero

Working title The source of extra nitrogen in Mrk996

Description Recently, JWST observations have unveiled a population of galaxies at high redshift with an excess in the nitrogen-to-oxygen relative abundance. Several mechanisms have been proposed to explain this excess, including Wolf-Rayet stars, supermassive stars, differential outflows, or an incomplete understanding of how the calibrations and methods used to derive abundances in the local Universe should be applied at high redshift. Looking at the heart of similar systems in the Local Universe, where we can resolve all the different galactic components (i.e., stars, gas, dust), can help us gain insight into what is causing this extra nitrogen. However, suitable examples for such an

AGREEMENT

	Name:	Signature:
Student	<u>Margarida Pólvara Fonseca</u>	<u>Margarida Pólvara Fonseca</u>
Supervisor 1	<u>Ana Monreal Ibero</u>	<u>Ana Monreal Ibero</u>
Supervisor 2	<u></u>	<u></u>
Studyadvisor	<u>Wouter Schrier</u>	<u></u>

TO DO

- 1) The student fills in the form together with his/her supervisor(s).
- 2) The student digitally hands in the filled in and signed form to the studyadvisor for approval: studyadvisor@strw.leidenuniv.nl
- 3) After approval, the studyadvisor sends a digital copy of the form to the student, supervisor(s) and the programme coordinator.
- 4) Upon completing the research project a grading form has to be filled in by the supervisor(s), this form can be found on http://www.strw.leidenuniv.nl/education/currentyear/Researchproject_Grading_form_Stk.pdf



APPENDIX REGISTRATION FORM MASTER RESEARCH PROJECT ASTRONOMY



Universiteit Leiden

RESEARCH PROJECTS

The student is expected to spend $30\text{EC} = 30 \times 28 = 840$ hours (1 EC equals 28 hours of work) on both the First Research Project and the Master's Research Project. The total credit includes 27 EC Research and 3 EC Thesis. For the Master's Research Project also a passing public presentation is included (the Student Colloquium). The First and Master's Research Projects must be on different topics. The Master's Research Project can be started only after successful completion of the First Research Project.

COLLOQUIUM

Every student should deliver a Student Colloquium. For the specialisations Research, Data Science, Cosmology, Instrumentation and Education, this colloquium should describe the Master's Research Project. For the specialisations Business Studies or Communication, the colloquium can describe either the Master's Research Project, or another research project or internship carried out in the context of the specialisation. You have to make your own reservation for the colloquium on the designated webpage: Sterrewacht homepage -> Local Pages -> Master Colloquium. Note: The colloquium can only be given on certain dates. The reservation is your own responsibility!

DEADLINES

The maximum duration of any Research Project is 9 months.

First Research Project

The deadline for the First Master Project is extended to **July 31, 23.59hrs** for students who started the programme in September, if the First Research Project is started before November 1 of the first year. For students who started the programme in February this deadline is extended to **December 31, 23.59hrs** if the First Research Project is started before April 1 of the first year.

Master Research Project

For the Master Research Project the deadline is **June 30, 23.59hrs** for students who started the programme in September and **November 30, 23.59hrs** for students who started the programme in February.

For all Research Projects, it is **not** possible to obtain a grade higher than 6 if the deadline is exceeded, unless the studyadvisor deems the delay to be caused by circumstances outside the control of the student.

GRADING

A master's thesis will be evaluated both by the supervisor(s) of the project and by another staff member not directly connected to the same research project (the second reader). Both the supervisor(s) and the second reader have to be approved staff members. The project will be evaluated on three points:

1. The quality and depth of the research
2. The dedication and initiative shown by the student
3. The quality of the thesis (and presentation)

PROGRESSION

To monitor the progression of the Research Projects you will receive periodic emails at so-called **milestones**. These milestones occur at the start of the project and at 25%, 50%, 75% and 100% of the allotted duration of the project. These emails contain information about the remaining time until the end date (at 100%). If the project has not been completed by the final date (100%), a grade will be given based on the available material. Student and supervisor should monitor progress with this schedule in mind. Problems or delays should be reported to the studyadvisor, who will also monitor progress.

The finished thesis has to be sent to the supervisor(s), the programme coordinator (eduster@strw.leidenuniv.nl) and the study advisor on the day of the final deadline at the latest!

ISSUES

Should personal problems, problems with your project or problems with your supervisor occur which you would prefer to discuss confidentially you can contact the studyadvisor Wouter Schrier (Oort 568),
studyadvisor@strw.leidenuniv.nl